# Data Collection & Initial Auditing Summary

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## 1. Dataset Overview & Source

The analysis uses the car_prices.csv dataset, which contains vehicle auction and sales transactions from 2014 to 2015. Each record represents a sold vehicle along with its technical specifications, condition, location, market value (MMR), and final selling price.

| Total Non-Empty Rows | Total Columns | Date Range | Median Price |
|---|---|---|---|
| **302,021** | 16 | 2014-01-01 to 2015-07-20 | ,800 |

## 2. Columns & Data Types

| Data Category | Column Names | Initial Audit Note |
|---|---|---|
| **Key / Identifier** | vin | Unique vehicle identification number; used to track repeated vehicles. |
| **Technical Specs** | make, model, trim, body, transmission, color, interior | Categorical fields describing car specs. Several need text cleaning. |
| **Numeric Metrics** | year, condition, odometer, mmr, sellingprice | Numeric values. MMR represents estimated market value, sellingprice is actual. |
| **Location & Date** | state, seller, saledate | saledate is stored as text string and must be parsed into date format. |

## 3. Data Quality Issues Identified

A detailed audit of the 302,021 non-empty rows revealed the following quality issues:

| Column | Missing Count | Missing % |
|---|---|---|
| **transmission** | 33,436 | 11.07% |
| **condition** | 11,793 | 3.90% |
| **body** | 7,645 | 2.53% |
| **trim** | 5,101 | 1.69% |
| **model** | 4,825 | 1.60% |
| **make** | 4,772 | 1.58% |
| **color / interior** | 461 | 0.15% |
| **odometer** | 81 | 0.03% |
| **sellingprice / mmr** | 0 | 0.00% (Clean) |

- Key Observations:

• Empty Rows in CSV: Found 256,174 completely blank rows at the end of the raw CSV file that must be dropped.

• Duplicates & Repeated VINs: Zero exact duplicate rows were found. However, 3,766 records share a repeated VIN, indicating cars that were sold more than once in the auction market.

• Outliers & Suspicious Values: Selling price ranges from  to ,500 (with 11 rows <=  representing clear data entry errors). Odometer includes 44 records at 500,000+ miles with some placeholders at 999,999.

• Date Formatting: saledate column contains full timestamp strings with timezone offsets (e.g., Tue Dec 16 2014 12:30:00 GMT-0800) which requires standard conversion to YYYY-MM-DD.

## 4. Handover to Data Cleaning Phase

Based on this audit, the dataset is handed over to the Data Cleaning team (Mai & Mohammed) with the following recommended action plan:

- 1. Text & Specs Cleaning (Mai): Standardize make/model/body casing, fill or categorize missing transmission as Unknown, and remove blank rows.
- 2. Numerical & Date Cleaning (Mohammed): Convert saledate to standard Date type, filter out invalid prices (< ), and verify extreme odometer readings.
- 3. Next Step: Export cleaned dataset to Excel and import into SQL for Data Modeling and KPI calculations.